Internet of Things (IoT) with Applications

Undergraduate Engineering Course

Module 2

IoT Communication and Networking

Short- and Long-Range Links, Application Protocols, and Network Concepts

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Unit II: IoT Communication and Networking

Concepts

Topics Covered:

Short-Range Links: BLE, Zigbee, RFID, NFC, Wi-Fi · Long-Range Links: LoRaWAN, NB-IoT, LTE-M, 5G
Application Protocols: MQTT, CoAP, HTTP/HTTPS, AMQP, WebSockets
Network Concepts: IPv4/IPv6, 6LoWPAN, WSNs, Gateways, Addressing & Routing

1 Introduction

Module 1 established the layered reference architecture that every IoT system is built around, with the **Network Layer** responsible for carrying data from the Perception Layer to processing and application services. This module opens that layer up in detail. No single communication technology suits every IoT deployment: a wearable heart-rate monitor, a smart streetlight, and an agricultural sensor scattered across a hundred-acre farm each face radically different constraints on power, range, data rate, and cost, and each is consequently served by a different point in the design space.

This module develops that design space along three axes. First, the physical- and data-link-layer **communication technologies** that move bits over the air, split into short-range and long-range families. Second, the **application-layer protocols** – MQTT, CoAP, HTTP/HTTPS, AMQP, and WebSockets – that structure the messages carried over those links. Third, the broader **network concepts** – addressing, routing, and the wireless sensor network topologies – that tie many devices together into a coherent deployment.

1.1 The Fundamental Trade-off: Range, Power, and Data Rate

Nearly every choice of IoT communication technology reduces to a three-way trade-off between **range** (how far a signal must travel), **power consumption** (how long a battery-powered device can operate unattended), and **data rate** (how much information can be moved per second). Technologies that achieve long range and low power typically do so by sacrificing data rate and latency; technologies that offer high data rate typically consume more power and cover less distance. Recognising which corner of this trade-off an application needs is the first step in any IoT network design, and is the organising principle behind Sections 2 and 3 below.

2 Short-Range Communication Technologies

Short-range technologies typically cover distances from a few centimetres up to around 100 metres, and are the natural choice within a single room, building, or vehicle.

2.1 Bluetooth and Bluetooth Low Energy (BLE)

Bluetooth provides moderate-range, moderate-data-rate wireless communication originally designed for streaming audio and peripheral connections. **Bluetooth Low Energy (BLE)**, a lighter-weight variant introduced with Bluetooth 4.0, sacrifices some data rate and range in exchange for dramatically lower power consumption, making it the dominant choice for battery-powered wearables and other IoT nodes that only need to send small amounts of data intermittently.

2.2 Zigbee

Zigbee is a low-power, low-data-rate protocol built on the IEEE 802.15.4 standard, designed specifically for **mesh networking**: devices can relay each other's messages, extending effective coverage well beyond the range of a single radio and allowing a network to self-heal if one node fails. This makes Zigbee well suited to smart home and building automation deployments with many fixed-position devices.

2.3 RFID and NFC

As introduced in Module 1, **RFID** and **NFC** are primarily identification technologies rather than general-purpose data links, but both are also communication technologies in their own right: RFID readers can interrogate many passive tags simultaneously over a range of centimetres to several metres, while NFC's much shorter range (a few centimetres) is itself a security feature, since it requires physical proximity to communicate.

2.4 Wi-Fi

Wi-Fi (IEEE 802.11) offers the highest data rate of the short-range family, sufficient for video streaming and large file transfer, at the cost of significantly higher power consumption than BLE or Zigbee. Wi-Fi is the natural choice for IoT nodes that are mains-powered or otherwise not battery-constrained, and where existing home or enterprise Wi-Fi infrastructure is already available.

Technology	Range	Data Rate	Power	Topology	Typical Use
BLE	∼10–30 m	Low	Very low	Star	Wearables, beacons
Zigbee	∼10–100 m	Low	Low	Mesh	Smart home, building automation
RFID	cm – a few m	Very low	None (passive)	Point-to-point	Asset tracking, access control
NFC	< 10 cm	Low	Very low	Point-to-point	Payments, pairing
Wi-Fi	∼30–100 m	High	High	Star/infra.	Cameras, gateways, high-throughput nodes

Table 1: Comparison of short-range IoT communication technologies.

Worked Example

Selecting a short-range technology. A building-wide occupancy-sensing deployment has 200 battery-powered motion sensors spread across five floors, each needing to report only an occasional binary event, with several sensors out of direct radio range of any central access point. Which short-range technology is the best fit, and why?

Solution. Zigbee is the best fit: its low power consumption suits long battery-powered operation for an occasional binary event, and, critically, its mesh topology allows sensors out of direct range of a central point to relay through neighbouring sensors – a capability BLE's star topology and Wi-Fi's higher power draw do not offer as naturally.

3 Long-Range Communication Technologies

When devices are spread over kilometres rather than metres – across a farm, a city, or a national utility network – short-range technologies cannot reach a gateway directly, even with mesh relaying. Long-range technologies solve this either by trading data rate for range at very low power (**LPWAN**, Low-Power Wide-Area Network, technologies), or by relying on cellular infrastructure.

3.1 LoRaWAN

LoRaWAN is an LPWAN technology capable of ranges up to several kilometres in open terrain, at very low power, but at a low data rate and with duty-cycle restrictions that limit how often a device may transmit. It operates over unlicensed spectrum, so deployments do not require a subscription to a cellular carrier, at the cost of managing the radio infrastructure (gateways) directly.

3.2 NB-IoT and LTE-M

NB-IoT (Narrowband IoT) and **LTE-M** are both cellular LPWAN technologies, operating over licensed spectrum using existing carrier infrastructure. NB-IoT is optimised for very low power and very low data rate, similar in profile to LoRaWAN but with the reliability and coverage guarantees of licensed cellular spectrum; LTE-M offers somewhat higher data rate and supports mobility (handover between cell towers) better than NB-IoT, making it more suitable for moving assets such as tracked vehicles.

3.3 4G/5G

Standard **4G** and **5G** cellular connectivity offer the highest data rate and lowest latency of the long-range family, at correspondingly higher power consumption. 5G's **ultra-reliable low-latency communication (URLLC)** mode is particularly significant for IoT: it targets millisecond-scale latency, enabling near-real-time control of remote actuators – a capability none of the LPWAN technologies can offer – making 5G the natural choice for latency-sensitive applications such as remote-controlled industrial robotics or connected-vehicle safety systems.

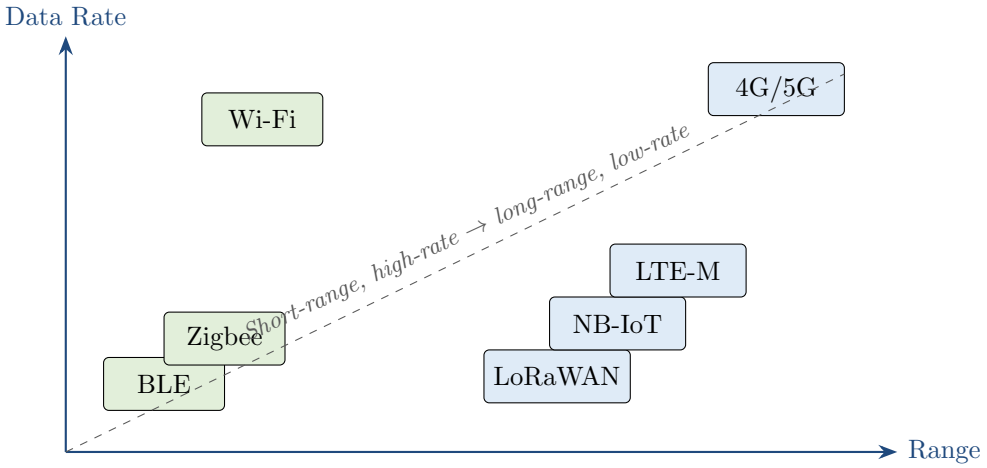


Figure 1: Range versus data-rate positioning of common short-range (green) and long-range (blue) IoT communication technologies. Power consumption broadly increases moving upward and to the right.

Worked Example

Selecting a long-range technology. A national water utility wants to read 50,000 static water meters, scattered across a large metropolitan area, once per day, with each meter expected to run on a single battery for at least eight years without unlicensed-band interference risk. Which long-range technology best fits, and why?

Technology	Range	Data Rate	Spectrum	Typical Use
LoRaWAN	2–15 km	Very low	Unlicensed	Agriculture, remote asset monitoring
NB-IoT	1–10 km	Very low	Licensed (cellular)	Smart meters, static sensors
LTE-M	1–10 km	Low–moderate	Licensed (cellular)	Tracked/mobile assets
4G/5G	Cell-tower dependent	High	Licensed (cellular)	Connected vehicles, low-latency control

Table 2: Comparison of long-range IoT communication technologies.

Solution. NB-IoT: it offers the very low power consumption needed for an eight-year, once-daily reporting schedule, licensed-spectrum reliability appropriate for a utility-scale deployment (avoiding unlicensed-band congestion risk), and does not need LTE-M's mobility support since the meters are static.

4 IoT Application Protocols

Once a physical link exists, IoT devices still need an agreed *message format* and *exchange pattern* for exchanging data. This is the role of the application-layer protocols below, each designed around different constraints on device resources and network reliability.

4.1 MQTT

MQTT (Message Queuing Telemetry Transport) is a lightweight **publish/subscribe** protocol designed specifically for constrained devices and unreliable networks. Devices **publish** messages to a named **topic** on a central **broker**; other devices or services **subscribe** to topics of interest and receive messages automatically whenever a publisher posts to them, without publisher and subscriber ever needing to know about each other directly. This decoupling is what makes MQTT the dominant protocol for large-scale sensor telemetry.

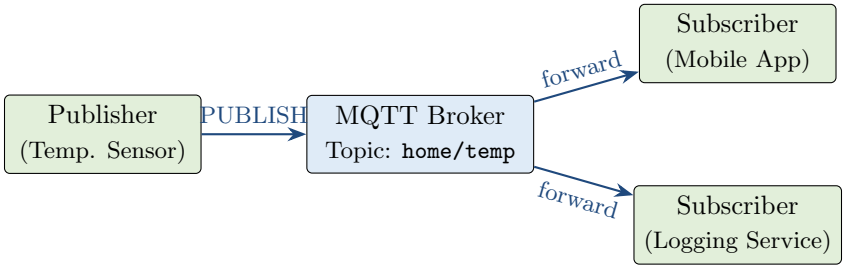


Figure 2: MQTT publish/subscribe messaging. A publisher sends data to a topic on the broker; the broker forwards each message to every subscriber of that topic, so publishers and subscribers remain fully decoupled.

4.2 CoAP

CoAP (Constrained Application Protocol) is a lightweight protocol modelled on HTTP's request/response pattern (GET, POST, PUT, DELETE) but redesigned to run over **UDP** rather than TCP, and to minimise message overhead – making it suitable for the most resource-constrained microcontrollers, where MQTT's persistent broker connection may itself be too heavy.

4.3 HTTP/HTTPS

HTTP and its encrypted form **HTTPS** are the familiar request/response protocols of the World Wide Web. They are simple to integrate with existing web infrastructure and tooling, but their comparatively large message overhead and synchronous, connection-per-request model make them less efficient than MQTT or CoAP for frequent, small telemetry messages from constrained devices.

4.4 AMQP

AMQP (Advanced Message Queuing Protocol) is a more feature-rich messaging protocol than MQTT, supporting complex routing, message queuing with guaranteed delivery, and transactional messaging. It is typically used on the *server side* of an IoT platform – moving messages reliably between backend services – rather than directly on constrained edge devices.

4.5 WebSockets

WebSockets provide a persistent, full-duplex connection over a single TCP connection, allowing a server to push data to a client (such as a browser-based dashboard) without the client needing to repeatedly poll. WebSockets are commonly used between an IoT cloud platform and a live dashboard, rather than between the edge device and the gateway.

Protocol	Pattern	Transport	Overhead	Typical Role
MQTT	Publish/subscribe	TCP	Very low	Constrained device telemetry
CoAP	Request/response	UDP	Very low	Ultra-constrained device control
HTTP/HTTPS	Request/response	TCP	High	Web integration, configuration APIs
AMQP	Queued messaging	TCP	Moderate–high	Backend/server-side message routing
WebSockets	Full-duplex stream	TCP	Low (after handshake)	Live dashboards, push updates

Table 3: Comparison of common IoT application-layer protocols.

Worked Example

Choosing a protocol. A fleet of 10,000 battery-powered soil sensors publishes a reading every 15 minutes to a cloud platform, which must in turn push live updates to a web-based farm dashboard viewed by several agronomists. Identify the most appropriate protocol for each leg of this pipeline.

Solution. Sensor-to-cloud: MQTT, since its publish/subscribe pattern and low overhead suit large numbers of constrained, battery-powered publishers reporting infrequently to a broker.

Cloud-to-dashboard: WebSockets, since the dashboard needs a persistent, low-latency, server-push connection rather than a decoupled topic-based model.

5 IoT Network Concepts

5.1 IPv4 and IPv6

IPv4, the long-standing addressing scheme of the Internet, provides approximately 4.3 billion unique addresses – a number that IoT's device scale can exhaust quickly. **IPv6** expands the address space to 2^{128} addresses, effectively unlimited for any foreseeable IoT deployment, and is consequently the addressing scheme of choice for large-scale IoT networks going forward, though many deployments still operate behind IPv4 gateways using private addressing internally.

5.2 6LoWPAN

6LoWPAN (IPv6 over Low-Power Wireless Personal Area Networks) is a standard that compresses and adapts IPv6 packets so they can be carried over low-power, low-data-rate links such as IEEE 802.15.4 (the same physical layer Zigbee is built on). It allows even severely resource-constrained devices to be addressed directly as first-class IPv6 hosts, rather than hidden entirely behind a translating gateway.

5.3 Wireless Sensor Networks (WSNs)

A **Wireless Sensor Network** is a collection of spatially distributed sensor nodes that cooperate to monitor a physical or environmental condition, typically organised in one of three topologies:

- **Star topology:** Every node communicates directly with a central coordinator; simple, but limited by the coordinator's radio range and represents a single point of failure.
- **Mesh topology:** Nodes relay messages for one another, extending effective range and adding redundancy, at the cost of more complex routing (as used by Zigbee, Section 2.2).
- **Tree topology:** A hybrid in which nodes are organised hierarchically under intermediate routers that themselves connect to a coordinator, balancing simplicity against some of the range extension of a full mesh.

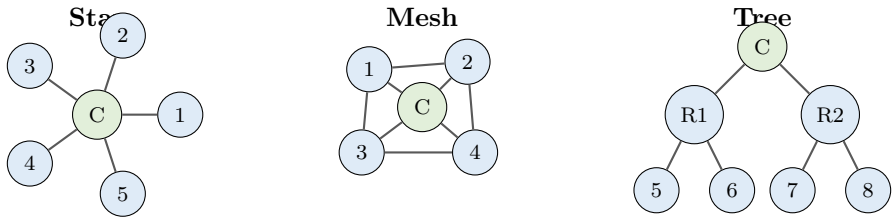


Figure 3: The three common Wireless Sensor Network topologies. C denotes the coordinator (gateway); in the tree topology, R1/R2 are intermediate routers.

5.4 Gateways, Addressing, and Routing

A **gateway**, first introduced in Module 1, sits between a local WSN and the wider Internet, typically performing protocol translation (e.g., 6LoWPAN/Zigbee to standard IP), aggregation, and

sometimes edge processing before forwarding data onward. **Addressing** assigns each node a unique identifier so that messages can be delivered to the correct destination, while **routing** determines the path a message takes across a multi-hop network such as a mesh – both are complicated in WSNs relative to traditional wired networks by node mobility, battery constraints (routing protocols must themselves consume minimal power), and frequent topology changes as nodes join, fail, or move.

IoT in Practice: Connecting Back to the Course

The technologies in this module slot directly into the Network Layer of the reference architecture from Module 1: short- and long-range radios and the WSN topologies of Section 5 implement the physical carriage of data, while MQTT, CoAP, and the other application-layer protocols of Section 4 structure the messages exchanged between the Perception and Processing Layers. Unit III builds directly on this foundation, showing how the ESP32, Arduino, and Raspberry Pi platforms implement these same protocols in code and how gateways hand data off to cloud and edge processing; Unit V's application domains each select a specific combination from this module – LoRaWAN and MQTT for smart agriculture, BLE and a cloud dashboard for wearable healthcare, NB-IoT for smart metering – matched to that domain's particular range, power, and latency requirements.

6 Solved Problems

Worked Example

Problem 1. A wildlife-tracking collar must report an animal's GPS location once every 30 minutes from remote, unpopulated terrain with no cellular coverage, and must run for at least two years on a small battery. Which long-range technology is most appropriate, and why would 4G/5G be a poor choice here?

Solution. LoRaWAN is most appropriate: it does not depend on cellular network coverage (which is absent in remote terrain) and its very low power consumption suits a two-year, infrequent-reporting battery budget. 4G/5G would be a poor choice both because cellular coverage is unavailable in the deployment area and because its high power draw would exhaust the battery far too quickly for infrequent, small GPS reports.

Worked Example

Problem 2. Explain why MQTT's publish/subscribe pattern scales better than a direct request/response model (such as plain HTTP polling) when 5,000 sensors must all report to 3 different backend services that each need every reading.

Solution. Under publish/subscribe, each sensor publishes once per reading regardless of how many subscribers exist, and the broker handles fan-out to all three backend services; under direct request/response polling, each of the 3 services would need to separately contact (or be contacted by) all 5,000 sensors, multiplying the number of connections and requests by the number of subscribers and coupling publishers directly to every consumer.

Worked Example

Problem 3. A single IPv4 address space (about 4.3×10^9 addresses) is shared globally across all Internet-connected devices, not just IoT. If a national smart metering rollout alone requires

100 million globally addressable devices, briefly explain why IPv6, not IPv4, is the appropriate addressing scheme, and name the technology that lets such devices use IPv6 over a low-power radio link.
Solution. IPv4's total address space is already heavily consumed by existing Internet hosts, so allocating 100 million more global IPv4 addresses to a single rollout is impractical; IPv6's 2^{128} address space comfortably accommodates this and far larger deployments. 6LoWPAN is the technology that compresses and carries IPv6 packets over low-power wireless links such as IEEE 802.15.4.

Worked Example

Problem 4. A smart-home network has one coordinator and twelve fixed sensor nodes, several of which are in rooms with poor direct radio line-of-sight to the coordinator. State which WSN topology is most appropriate and justify your choice in terms of the specific limitation of a star topology it overcomes.
Solution. A mesh (or tree) topology is most appropriate. A star topology requires every node to reach the coordinator directly, so nodes with poor line-of-sight would drop out of the network entirely; a mesh allows such nodes to relay through neighbouring nodes that do have a clear path to the coordinator, extending effective coverage without adding radio power.

7 Summary

The Network Layer of an IoT deployment is chosen along a persistent trade-off between range, power consumption, and data rate. Short-range technologies – BLE, Zigbee, RFID, NFC, and Wi-Fi – cover single buildings or rooms, with Zigbee's mesh topology and BLE's low power suiting most battery-constrained fixed deployments and Wi-Fi suiting higher-throughput, mains-powered nodes. Long-range technologies – LoRaWAN, NB-IoT, LTE-M, and 4G/5G – extend coverage to kilometres, trading data rate for range and power at the LPWAN end, and offering high data rate and low latency at the cellular end. Above these physical links, application-layer protocols such as MQTT and CoAP structure lightweight telemetry from constrained devices, while HTTP/HTTPS, AMQP, and WebSockets serve integration, backend-routing, and live-dashboard roles respectively. Finally, IPv6 and 6LoWPAN address the scale of modern IoT deployments directly at the IP layer, and wireless sensor network topologies – star, mesh, and tree – together with gateways, addressing, and routing, determine how many devices can be tied together into a single coherent network.

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